# AARAV ARYA

B.Tech Electrical Engineering — IIT Mandi
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## Education

- **Indian Institute of Technology, Mandi** — Mandi, Himachal Pradesh, India
  *Bachelor of Technology - Electrical Engineering;* ***CGPA: 9.2/10*** — 2026 – Present
  - *Relevant Coursework:* Calculus I, Complex & Vector Calculus, Linear Algebra, ODE & Integral Transforms, Probability & Statistics, Programming & Data Structures
- **Ramseth Thakur Public School, Kharghar** — Navi Mumbai, Maharashtra, India
  *Class XII (CBSE);* ***Score: 96%*** — 2025
- **Delhi Public School, Nerul** — Navi Mumbai, Maharashtra, India
  *Class X (CBSE);* ***Score: 97.6%*** *(Top 5 Subjects)* — 2023

## Technical Skills

**Languages:** Python, C++,Arduino, JavaScript, HTML/CSS
**ML/AI:** PyTorch, CUDA, NumPy, Matplotlib, Scikit-learn, Ollama, DINOv2, DeepLabV3
**Tools:** GitHub, VS Code, Jupyter Notebooks, Arduino IDE, ANACONDA,Sublime text

## Projects

**Off-Road Semantic Segmentation — KrackHack 2026**

- Built a terrain perception pipeline for autonomous off-road navigation using a DINOv2 (ViT) backbone with a DeepLabV3-inspired decoder incorporating ASPP for multi-scale feature extraction
- Improved segmentation performance from **0.28 to 0.54 mIoU** (+0.26 absolute, ∼93% relative gain) through **5 systematic iterations** involving architecture refinement, hyperparameter tuning, and error analysis
- Mitigated class imbalance across terrain classes using weighted loss functions, improving segmentation quality on underrepresented regions
- Accelerated experimentation using parallelized multi-GPU training, reducing iteration time and enabling rapid hypothesis validation
- Evaluated model robustness across diverse terrain types, lighting conditions, and slope variations, ensuring consistent performance in unstructured environments

**VeriSync — GDG 2025 (Pitch) — Concept**

- Conceptualized a decentralized credential verification system leveraging Soulbound NFTs to ensure non-transferable, tamper-proof academic and professional records
- Defined trust model enabling universities to issue credentials, students to own them, and employers to perform instant, trustless verification without intermediaries
- Analyzed limitations of traditional verification systems (manual processes, forgery risk, latency) and proposed blockchain-based solutions to eliminate credential fraud and reduce verification time
- Outlined scalable workflow for credential issuance, storage, and verification, including metadata handling and on-chain/off-chain data separation for efficiency

**MNIST Digit Classifier**

- Built a CNN-based handwritten digit classifier trained on the MNIST dataset (60,000 training, 10,000 test samples) achieving **99.12% test accuracy**
- Designed and tuned convolutional architecture with multiple layers, optimizing kernel sizes, activation functions, and regularization techniques to improve generalization
- Achieved rapid convergence within a few epochs ($< 10$), maintaining low validation loss and minimal overfitting
- Evaluated model performance using accuracy and loss metrics, consistently achieving $< 1\%$ classification error across test set
- Integrated real-time training visualization using **Matplotlib** to track loss curves and model convergence behavior across epochs

**Systematic Portfolio Tracker & Transaction Engine**

- Developed a multi-strategy paper-trading engine in Python tracking 6 parallel models (Trend-Following, Mean-Reversion) on Indian Blue-Chips, benchmarked dynamically against the Nifty 50 index
- Re-engineered the state machine to use a transactional SQLite database with write-ahead logging (WAL) for ACID compliance, eliminating desynchronization and double-buy risks during API drops
- Built a responsive dark-mode performance dashboard using HTML, JavaScript, and Chart.js featuring daily PnL charting, synchronized calendar timelines, and portfolio metrics filtering

## Competitions & Achievements

- **Anatolian Rover Challenge (ARC) 2026 — Global Rank 12 (Finalist)**; qualified as the sole Indian team for the finals in Ankara, Turkey with Team Deimos; contributed to rover system re-engineering, life sciences experimentation, and led sponsorship logistics and budget planning
- **Robocon ABU (IIT Mandi) — Ranked 12/77 nationally**; contributed to CAD design of tri-wheel locomotion system, simulation/animation for submissions, and assisted in mechanical prototyping and validation
- **Google GDG (Web3)** — Presented VeriSync system architecture, outlining decentralized credential verification using a blockchain-based trust model
- **Robowar Winner — Xpecto, IIT Mandi** — Led electrical and control subsystem development; designed CAD for weapon motor mount and **integrated 2 DC motors, 2 encoders, and an IMU with a Teensy 4.0 microcontroller via motor driver**, implementing wiring, power distribution, and control logic for real-time actuation
- **Codeforces** — Achieved a competitive programming rating of **1207** (Profile)

## Extracurricular

- **Robotronics Club — IIT Mandi** — Core Member; part of Publicity & Management Team
- **Kulhadd Fest** — Initiated outreach, pitched and led negotiations to **secure PNB sponsorship**
- **GriffinSight Club** — Media coverage and event photography